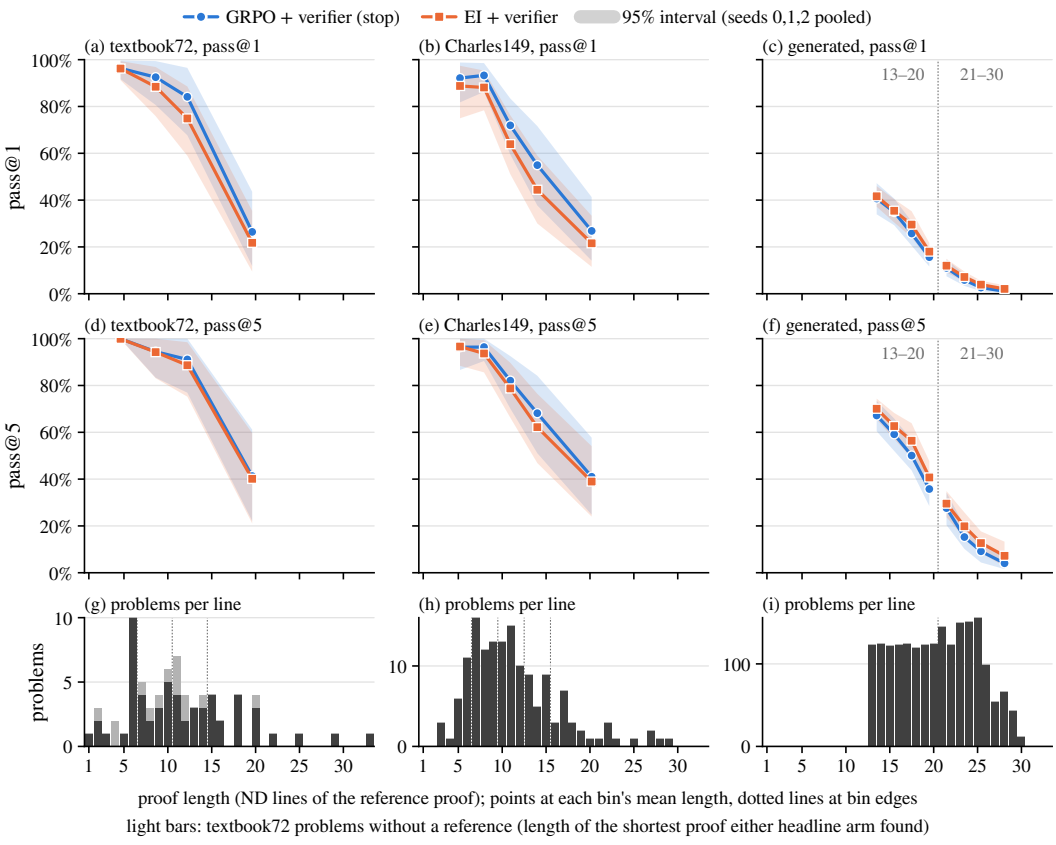
GRPO + verifier (stop)
EI + verifier
95% interval (seeds 0,1,2 pooled)
(a) textbook72, pass@1
(b) Charles149, pass@1
(c) generated, pass@1
13–20
21–30
pass@1
100%
80%
60%
40%
20%
0%
(d) textbook72, pass@5
(e) Charles149, pass@5
(f) generated, pass@5
pass@5
(g) problems per line
(h) problems per line
(i) problems per line
problems
1
5
10
15
20
25
30
100
proof length (ND lines of the reference proof); points at each bin's mean length, dotted lines at bin edges
light bars: textbook72 problems without a reference (length of the shortest proof either headline arm found)